

Data Quality

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Why IS Data Quality Relevant?

- The consequences of poor quality of information can be experienced in everyday life but, often, without making explicit connections to their causes
 - Example: the late or mistaken delivery of a letter is often blamed on a dysfunctional postal service, although a closer look often reveals data-related causes, e.g., an error in the address, which can be traced back to the originating database.
 - Example: the duplicate delivery of automatically generated e-mails is often indicative of a database record duplication error.

Why is Data Quality Relevant?

- Information quality seriously impacts on the efficiency and effectiveness of organizations and businesses.
- The report on information quality of the Data Warehousing Institute [1] estimates that IQ problems cost US businesses more than 600 billion dollars a year.
- The findings of the report were based on interviews with industry experts, leading-edge customers, and survey data from 647 respondents.

[1] <http://www.dw-institute.com/>

Examples of the Importance of Data Quality in Organizational Processes

- Customer Matching: In private organizations, such as marketing firms or banks, it is not surprising to have several (sometimes dozens!) customer registries, updated by different organizational procedures, resulting in inconsistent, duplicate information.
- It is very difficult, e.g., for banks to provide clients with a unique list of all their accounts and funds.

Examples of the Importance of Data Quality in Organizational Processes

- Corporate householding. Many organizations establish separate relationships with single members of households or, more generally, related groups of people; either way, they like, for marketing purposes, to reconstruct the household relationships in order to carry out more effective marketing strategies.
- This problem is even more complex than the previous one, since in that case, the information to match concerned the same person, while in this case, it concerns groups of persons corresponding to the same household.

Examples of the Importance of Data Quality in Organizational Processes

- Organization fusion. When different organizations (or different units of an organization) merge, it is necessary to integrate their legacy information systems.
- Such integration requires compatibility and interoperability at any layer of the information system, with the database level required to ensure both physical and semantic interoperability.

The examples above are indicative of the growing need to integrate information across completely different data sources, an activity in which poor quality hampers integration efforts.

Fields that investigated Data Quality

- Information quality has been addressed in different areas, including **statistics, management, and computer science**. Statisticians were the first to investigate some of the problems related to information quality, by proposing a mathematical theory for considering duplicates in statistical datasets in the late 1960s.
- They were followed by researchers in management, who at the beginning of the 1980s focused on how to control information manufacturing systems in order to detect and eliminate information quality problems.
- Only at the beginning of the 1990s computer scientists began considering the problem of defining, measuring, and improving the quality of electronic information stored in databases, data warehouses, and legacy systems.

Data Quality

- When people think about data quality, they often reduce quality just to **accuracy**. For example, let us consider the surname “Belhajjame”; when this is spelled during a telephone call, several misspellings are reported by the other side, such as “Velhajjame,” “Belhajjane,” “Belajjame,” etc. all inaccurate versions of the original.
- Indeed, information is normally considered to be of poor quality if typos are present or wrong values are associated with a concept instance, such as an erroneous birth date or age associated with a person.
- However, information quality is more than simply accuracy. Other significant dimensions such as **completeness**, **consistency**, and **currency** are necessary in order to fully characterize the quality of information.

Data Quality Problems Illustrated

The table below provides some examples of these dimensions for structured data. It describes movies, with title, director, year of production, number of remakes, and year of the last remake.

Id	Title	Director	Year	#Remakes	LastRemakeYear
1	Casablanca	Weir	1942	3	1940
2	Dead poets society	Curtiz	1989	0	NULL
3	Rman Holiday	Wylder	1953	0	NULL
4	Sabrina	null	1964	0	1985

Data Quality Problems Illustrated

There is a misspelling in the title of movie 3, where Rman stands for Roman, thus causing an accuracy problem.

Id	Title	Director	Year	#Remakes	LastRemakeYear
1	Casablanca	Weir	1942	3	1940
2	Dead poets society	Curtiz	1989	0	NULL
3	Rman Holiday	Wylder	1953	0	NULL
4	Sabrina	null	1964	0	1985

Data Quality Problems Illustrated

Another accuracy problem is related to the exchange of the director between movies 1 and 2; Weir is actually the director of movie 2 and Curtiz the director of movie 1.

Id	Title	Director	Year	#Remakes	LastRemakeYear
1	Casablanca	Weir	1942	3	1940
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Data Quality Problems Illustrated

Other data quality problems are a missing value for the director of movie 4, causing a completeness problem, and a 0 value for the number of remakes of movie 4, causing a currency problem because a remake of the movie has actually been proposed.

Id	Title	Director	Year	#Remakes	LastRemakeYear
1	Casablanca	Weir	1942	3	1940
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3	Rman Holiday	Wylder	1953	0	NULL
4	Sabrina	null	1964	0	1985

Data Quality Problems Illustrated

The previous examples of dimensions concern the quality of data values represented in the relation.

Besides data, a large part of the design methodologies for the relational model addresses properties that concern the quality of the schema

For example, several normal forms have been proposed with the aim of capturing the concept of good relational schema, free of anomalies and redundancies.

Data Quality Problems Illustrated

Finally, there are two consistency problems: first, for movie 1, the value of LastRemakeYear cannot be lower than Year; second, for movie 4, the value of LastRemakeYear cannot be different from null, because the value of #Remakes is 0.

Id	Title	Director	Year	#Remakes	LastRemakeYear
1	Casablanca	Weir	1942	3	1940
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Issues with Data Quality

The previous examples and considerations show that:

- Data quality is a multifaceted concept, to the definition of which different dimensions concur
- Quality dimensions, such as accuracy, can be easily detected in some cases (e.g., misspellings) but are more difficult to detect in other cases (e.g., where admissible but not correct values are provided)
- A simple example of a completeness error has been shown, but like what happens with accuracy, completeness can also be very difficult to evaluate (e.g., if a tuple representing a movie is entirely missing from the relation Movie)
- Consistency detection does not always localize the errors (e.g., for movie 1, the value of the LastRemakeYear attribute is wrong)